

Multi-head Latent Attention

t	1	2	3	4	5	6
3	3	0	3	0		2
2	0	3	2	3		1
1	0	1	1	1		3
3	1	2	3	1		1
2	3	2	2	1		2
	↓	↓	↓	↓		↓

Model Size 5	<u>5</u>	7168
Context Length	<u>6</u>	
# of heads	<u>3</u>	128
per-head dimensions	<u>4</u>	128

Latent QKV

W ⁰ DQ	C ₁ ⁰ Q					
1 -1 1 0 1	4	6	0	4	-1	6
0 0 -1 1 0	2	1	1	2	0	-2
1 0 -1 0 1	4	6	1	4	0	1

W ⁰ DKV	C ₁ ⁰ KV					
1 -1 1 0 0	2	3	-2	2	-2	4
0 0 -1 0 0	-1	0	-1	-1	-1	-3

Compression Dimensions	
query	3
key/value	2

RoPE (Key)

W*KR											
1	-1	1	1	0	0	5	4	0	5	-1	5
0	0	-1	0	1	0	1	3	1	1	0	-1

R1	R2	R3	R4	R5	R6	K*R					
0.85	-0.5	0.43	-0.9	-0.1	-1	-0.6	-0.8	-0.9	-0.3	-1	0.23
0.53	0.85	0.9	0.43	0.99	-0.1	0.78	-0.6	0.32	-0.9	-0.2	-1

RoPE (Query)

W*QR										
0	0	2	1	1	14	19	2	14	-2	11
1	0	-1	1	-1	-6	-11	0	-6	1	-9

R1	R2	R3	R4	R5	R6	Q^R					
0.85	-0.5	0.43	-0.9	-0.1	-1	-0.6	-0.8	-0.9	-0.3	-1	0.23
0.53	0.85	0.9	0.43	0.99	-0.1	0.78	-0.6	0.32	-0.9	-0.2	-1

Head 1

Linear

Wq

-1	1	1	0	0	→
0	0	1	0	1	→
1	0	0	0	2	→
0	0	1	1	1	→

Q°C

4	6	0	4	-1	6
8	12	1	8	-1	7
8	12	2	8	0	2
10	13	2	10	-1	5

Wk

1	0	1	1	0	→
0	-1	1	1	1	→
1	0	2	1	1	→
0	1	1	1	1	→

K°C

2	3	-2	2	-2	4
1	3	-3	1	-3	1
1	3	-3	1	-3	1
1	3	-3	1	-3	1

Wv

-1	0	0	1	0	→
0	-1	1	0	1	→
1	1	0	1	1	→
0	0	1	1	1	→

V°C

2	3	-2	2	-2	4
-1	0	-1	-1	-1	-3
1	3	-3	1	-3	1
1	3	-3	1	-3	1

Scaled Dot-Product

Scaled Dot-Product

						Q ^R					
						Q ^C					
						K ^T * Q / sqrt(dk)					
						V ^C					
						</					

dk=6

Softmax

	1	2	3	4	5	6
1	0.99	0.01	0	0	0	0
2	0.01	0.99	0.99	1	0	0.46
3	0	0	0	0	0.21	0
4	0	0	0.01	0	0	0.02
5	0	0	0	0	0.78	0
6	0	0	0	0	0	0.52

V

	↓	↓	↓	↓	↓	↓
→	2.01	2.99	2.99	3	-2	3.5
→	-1	-0	-0	-0	-1	-1.6
→	1.02	2.98	2.98	3	-3	1.92
→	1.02	2.98	2.98	3	-3	1.92

Head 2

Linear

Wq						Q°C					
0	1	1	0	0	→	4	6	0	4	-1	6
0	1	1	0	1	→	8	12	1	8	-1	7
1	0	0	0	1	→	4	6	1	4	0	1
0	0	1	1	1	→	10	13	2	10	-1	5
Wk						K°C					
0	0	1	1	1	→	1	3	-3	1	-3	1
0	-1	1	0	1	→	-1	0	-1	-1	-1	-3
1	1	0	1	0	→	2	3	-2	2	-2	4
0	1	2	1	1	→	1	3	-3	1	-3	1
Ww						V°C					
-1	0	0	1	0	→	2	3	-2	2	-2	4
0	-1	1	0	1	→	-1	0	-1	-1	-1	-3
1	1	0	1	1	→	1	3	-3	1	-3	1
0	0	1	1	1	→	1	3	-3	1	-3	1

Scaled Dot-Product

[illegible]

	1	2	3	4	5	6
1	1	0.99	0.02	0	0.02	0
2	0	0.01	0.95	1	0	0.83
3	0	0	0	0	0.21	0
4	0	0	0.03	0	0	0.17
5	0	0	0	0	0.77	0
6	0	0	0	0	0	0

V	↓	↓	↓	↓	↓	↓
→	2	2.01	2.96	3	-1.9	2.83
→	-1	-1	-0.1	-0	-1	-0.2
→	1	1.01	2.9	3	-2.9	2.67
→	1	1.01	2.9	3	-2.9	2.67

Head 3

Linear

Wq						Q°C					
0	1	1	0	0	→	4	6	0	4	-1	6
0	0	0	0	1	→	4	6	1	4	0	1
1	0	0	0	1	→	4	6	1	4	0	1
0	1	0	1	1	→	6	7	2	6	0	-1

Wk						K°C					
0	1	1	1	1	→	1	3	-3	1	-3	1
0	-1	1	0	1	→	-1	0	-1	-1	-1	-3
1	1	0	1	0	→	2	3	-2	2	-2	4
1	2	1	1	1	→	1	3	-3	1	-3	1

Ww						V°C					
-1	1	0	1	0	→	2	3	-2	2	-2	4
0	1	0	0	0	→	0	0	0	0	0	0

Scaled Dot-Product

[illegible]

	1	2	3	4	5	6
1	1	1	0.02	0	0.07	0
2	0	0	0.95	1	0	0
3	0	0	0	0	0.2	0
4	0	0	0.03	0	0	0.82
5	0	0	0	0	0.73	0
6	0	0	0	0	0	0.18

1	1	1	1	1	→
0	0	0	1	0	→

1	3	-3	1	-3	1
2	3	-2	2	-2	4

V°C						
2	3	-2	2	-2	4	
0	0	0	0	0	0	
1	3	-3	1	-3	1	
2	3	-2	2	-2	4	

V'	↓	↓	↓	↓	↓	↓
→	2	2	2.96	3	-1.7	2.36
→	0	0	0	0	0	0
→	1	1	2.9	3	-2.7	1.01
→	2	2	2.96	3	-1.7	2.36

Concat

Head	V
1	2.010014 2.9917231 2.99 3 -1.987401867 3.502675549 -0.98999 -0.008277 -0.01 -0 -1.000088662 -1.579778456 1.020028 2.9834462 2.98 3 -2.987491729 1.922897093 1.020028 2.9834462 2.98 3 -2.987491729 1.922897093
2	2.000015 2.006137 2.96 3 -1.93021996 2.83434149 -0.99999 -0.993386 -0.05 -0 -1.002511203 -0.16672884 1.000029 1.0132273 2.9 3 -2.932731163 2.66761265 1.000029 1.0132273 2.9 3 -2.932731163 2.66761265
3	2 2.0000043 2.96 3 -1.682841388 2.35743542 0 0 0 0 0 0 1 1.0000086 2.9 3 -2.687519875 1.00530361 2 2.0000043 2.96 3 -1.682841388 2.35743542

Output Projection

1	0	0	0	1	1	1	0	1	0	0	0	→
1	1	1	0	0	0	1	0	0	1	1	0	→
0	1	0	0	0	1	0	0	0	1	0	0	→
1	1	0	1	0	1	1	1	1	0	0	1	→
0	1	0	1	0	1	1	1	0	1	0	1	→

	↓	↓	↓	↓	↓	↓
Output	6.010073	7.018182	11.8	12	-9.53570558	11.19533627
	4.040086	7.9801283	11.8	12	-11.59523449	7.518710447
	-1.98997	-1.001663	-0.06	-0	-2.002601065	-1.746507295
	7.04013	10.999969	17.6	18	-16.20863976	13.72916149
	3.030116	6.0082419	11.7	12	-12.53839651	7.869050519

Output

Mixture of Experts

	t=	1	2	3	4	5	6
Input		6.010073	7.018182	11.8	12	-9.53570558	11.19533627
		4.040086	7.9801283	11.8	12	-11.59523449	7.518710447
		-1.98997	-1.001663	-0.06	-0	-2.002601065	-1.746507295
		7.04013	10.999969	17.6	18	-16.20863976	13.72916149
		3.030116	6.0082419	11.7	12	-12.53839651	7.869050519
		1	1	1	1	1	1

of Experts 4
of shared 1

Router

Shared	E1												
Routed	E2	0	1	0	1	1	-10	4.110333	14.98834	31.1	32	-50.34227076	19.11692245
	E3	1	0	0	0	1	-20	-10.9598	-6.973576	3.43	4	-42.07410209	-0.935613211
	E4	1	1	0	0	0	-10	0.050159	4.9983103	13.5	14	-31.13094007	8.714046717

Sigmoid	1	1	1	1	1	1
0.98	1	1	1	1	1.4E-22	1
0	0	0.97	0.98	5.3E-19	0.28	
0.51	0.99	1	1	3E-14	1	

Rank	Sort by Columns	1	1	1	1	1	1
1	0.98386	0.999999691	1	1	0	1	
2	0.51254	0.993295907	1	1	0	1	
3	1.7E-05	0.000935423	0.97	0.98	0	0.28	

Gates	Top3	1	1	1	1	1	1
E1	0.98	1	1	1	0	1	
E2	0	0	0	0	0	0	
E3	0.51	0.99	1	1	0	1	

Expert 1

Linear	W	b											
1	0	1	0	-1	-3	-2.01001	-2.991723	-2.99	-3	-1.999910138	-1.420221544		
0	0	0	1	0	-2	-0.95985	3.0065787	9.61	10	-16.54099757	4.122543224		
1	-1	1	1	0	-1	6.020146	8.0363598	16.5	17	-17.15171191	14.65928002		
1	-1	1	1	1	-5	5.050262	10.044602	24.2	25	-33.69010842	18.52833053		
0	1	-1	-1	1	-20	-17.98	-16.00994	-14.1	-14	-25.92239018	-16.59489323		
0	1	0	0	1	-2	5.070202	11.98837	21.4	22	-26.133631	13.38776097		
1	1	1	-1	0	-25	-23.9799	-22.00332	-19.2	-19	-31.92490138	-21.76162207		
ReLU						0	0	0	0		0		
						0	3.0065787	9.61	10		4.122543224		
						6.020146	8.0363598	16.5	17		14.65928002		
						5.050262	10.044602	24.2	25		18.52833053		
						0	0	0	0		0		
						5.070202	11.98837	21.4	22		13.38776097		
						0	0	0	0		0		
Linear	W	b											
0	0	0	0	1	1	0	1						
0	0	0	1	1	1	-1	-2						
0	1	-1	1	1	0	0	1						
1	1	0	0	0	0	0	-1						
0	0	0	1	0	1	0	5						
ReLU						6.070202	12.98837	22.4	23		1	14.38776097	
						12.12046	24.032972	47.7	49		2	33.9160915	
						0.030116	6.0148206	18.3	19		1	8.991593743	
						-1	2.0065787	8.61	9		-1	3.122543224	
						15.12046	27.032972	50.7	52		5	36.9160915	
Gates	E1	1	1	1	1	1	1						
						6.070202	12.98837	22.4	23		1	14.38776097	
						12.12046	24.032972	47.7	49		2	33.9160915	
						0.030116	6.0148206	18.3	19		1	8.991593743	
						0	2.0065787	8.61	9		0	3.122543224	
						15.12046	27.032972	50.7	52		5	36.9160915	

Expert 2

Linear	W	b											
1	0	0	0	0	-3	3.01	4.02	8.75	9	-13	8.2		
1	0	0	-1	0	2	0.97	-2	-3.9	-4	8.67	-0.5		
1	1	1	1	0	2	17.1	27	43.1	44	-37	32.7		
0	1	1	1	1	1	13.1	25	42	43	-41	28.4		
0	0	0	1	1	2	12.1	19	31.3	32	-27	23.6		
0	1	1	1	0	-1	8.09	17	28.3	29	-31	18.5		
0	-1	0	0	0	3	-1	-5	-8.8	-9	14.6	-4.5		
ReLU						3.01	4.02	8.75	9	0	8.2		
						0.97	0	0	0	8.67	0		
						17.1	27	43.1	44	0	32.7		
						13.1	25	42	43	0	28.4		
						12.1	19	31.3	32	0	23.6		
						8.09	17	28.3	29	0	18.5		
						0	0	0	0	14.6	0		
Linear	W	b											
0	1	1	-1	0	1	1	-3						
1	0	1	0	-1	0	0	-2						
0	0	1	1	0	0	-1	-1						
0	0	1	0	0	0	1	-3						
1	0	-1	0	2	0	0	2						
ReLU						10	16	26.4	27	20.3	19.8		
						6.04	10	18.5	19	-2	15.3		
						29.2	51	84.1	86	-16	60.1		
						14.1	24	40.1	41	11.6	29.7		
						12.1	17	30.3	31	2	24.7		
Gates	E2	0.98	1	1	1	0	1						
						9.88	16	26.4	27	0	19.8		
						5.94	10	18.5	19	0	15.3		
						28.7	51	84.1	86	0	60.1		
						13.9	24	40.1	41	0	29.7		
						11.9	17	30.3	31	0	24.7		

Expert 3

Linear														
W						b								
1	0	0	0	0	0	-3	3.01007	4.02	8.75	9	-13	8.2		
1	2	0	-1	0	0	2	9.05012	14	19.7	20	-15	14.5		
1	1	1	1	0	0	2	17.1003	27	43.1	44	-37	32.7		
0	1	0	1	1	1	1	15.1103	26	42.1	43	-39	30.1		
0	0	0	1	1	1	2	12.0702	19	31.3	32	-27	23.6		
0	1	0	1	1	1	-1	13.1103	24	40.1	41	-41	28.1		
1	-1	0	0	0	0	3	4.96999	2.04	2.99	3	5.06	6.68		
ReLU														